MUSCLE ACTIVITY AND FATIGUE

Brain–Machine Interfaces

ABSTRACT

Learning how to use the Vernier hand dynamometer and EMG sensor, and understanding the signals obtained.

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1. Background for the Experiment

In this laboratory session, a hand dynamometer and an ECG/EMG sensor were used. The hand dynamometer records grip strength using a sensor that converts the applied force into a voltage. This voltage was recorded and then converted back into force in MATLAB using the linear calibration provided in the lab script:

$$\textbf{Force (N)} = 175.416 \cdot V - 19.296$$

In addition, three electrodes placed on a student's forearm were used to record the surface electromyogram (EMG), i.e. the electrical activity of the muscle. This signal corresponds to the sum of the motor unit action potentials generated during muscle contraction.

Both Vernier sensors (hand dynamometer and ECG/EMG sensor) were connected to an NI ELVIS II Series board through Analog Proto Board Connectors.

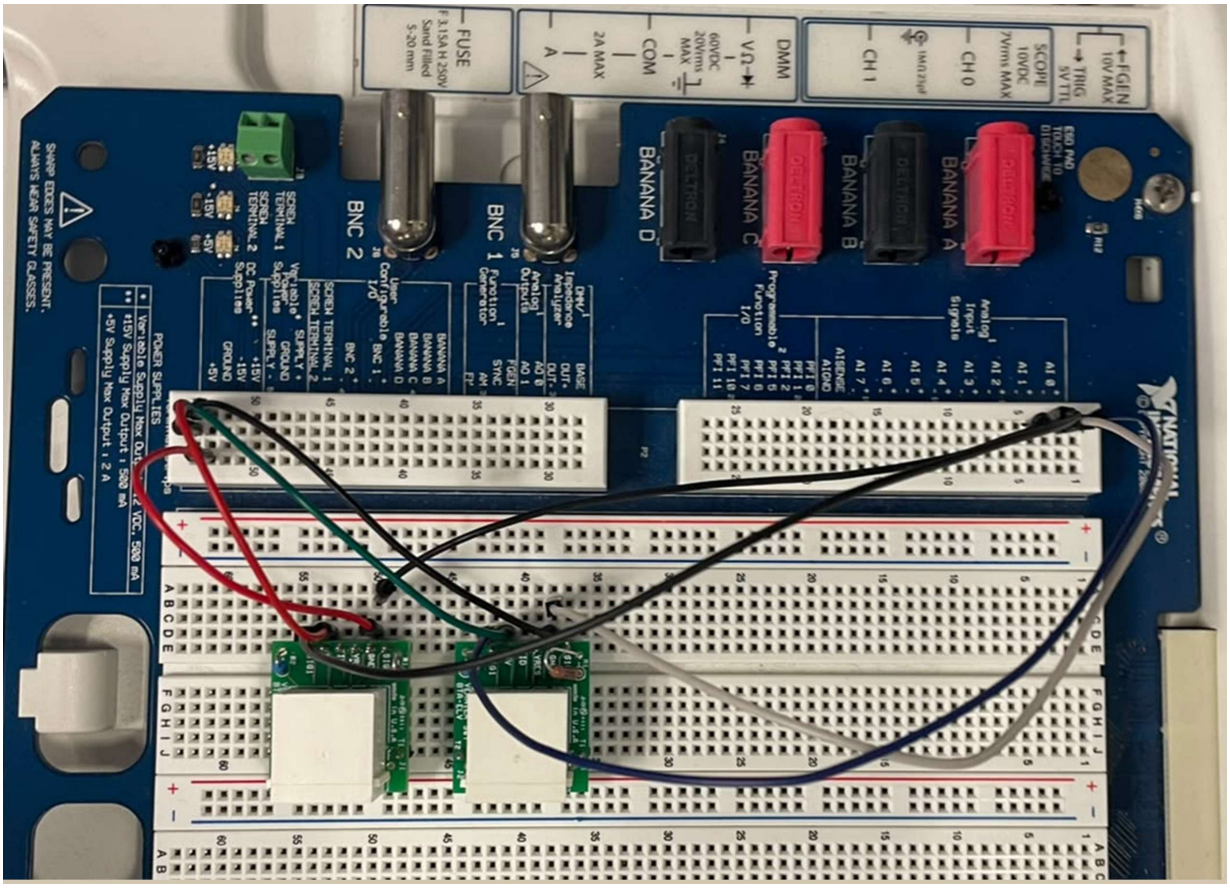


Figure 1. Vernier sensors connected to the NI ELVIS II board through Analog Proto Board Connectors.

1.1 Differences between the 'Force vs Time' and 'EMG vs Time' signals

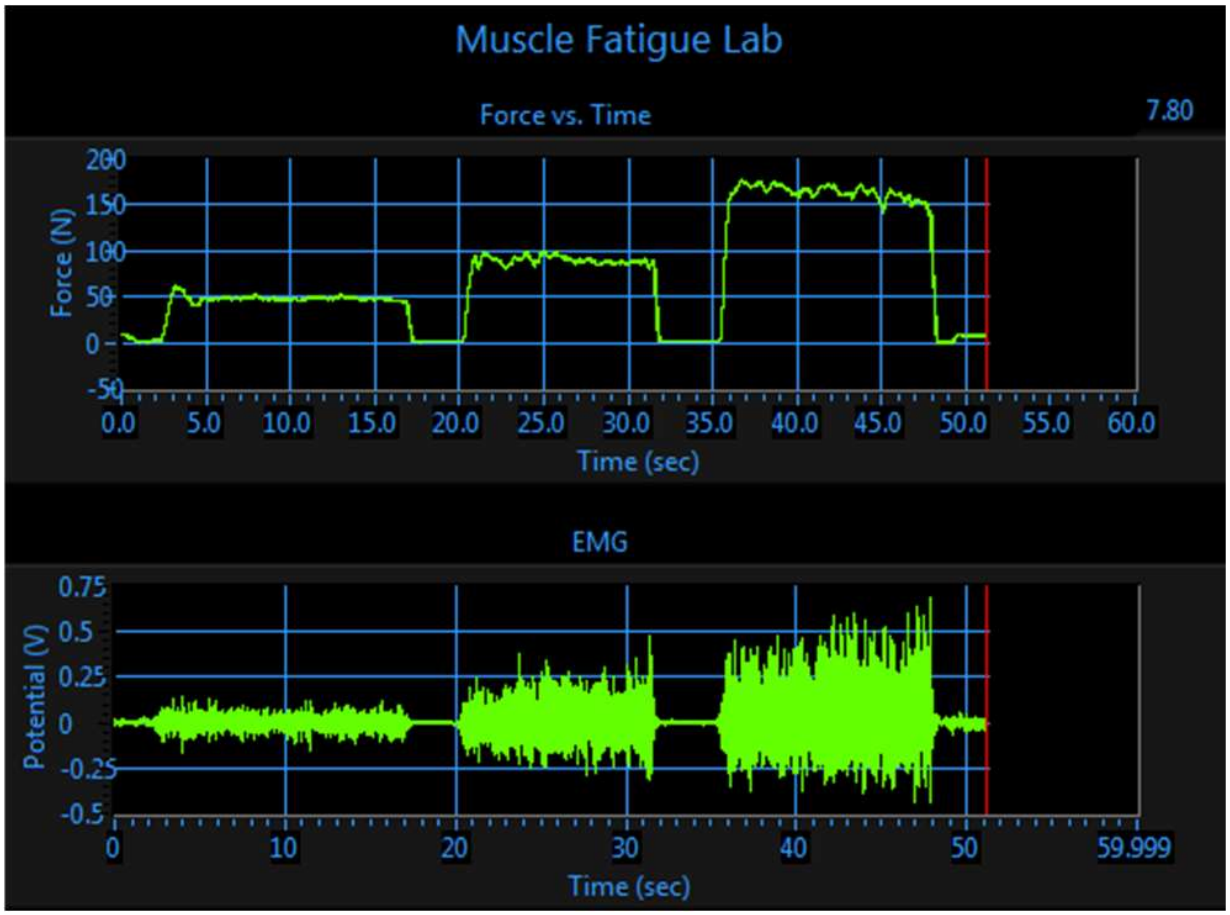


Figure 2. Reference figure from the lab script (Force and EMG vs Time).

Force and EMG are two different physical quantities. The force signal is obtained with strain-gauge technology: it is a smooth, low-frequency signal, and the output voltage is proportional to the applied force, so the voltage rises when the student squeezes the device. This voltage is then converted into newtons using the linear calibration of the device.

The EMG signal, on the other hand, is recorded by three electrodes placed on the forearm and picks up the action potentials of the muscle fibres. Its relationship with the applied force should also be roughly proportional (more force means more motor units recruited and/or higher firing rates, and therefore a larger and denser EMG signal). However, this relationship is a general tendency rather than a clean signal that rises and falls exactly with force, because the raw EMG is a noisy, high-frequency, zero-mean signal.

2. Grip Strength and Muscle Activity

In this first experiment, the grip strength of each student was recorded with the hand dynamometer and the EMG sensor simultaneously, following the timing and grip pattern in Table 1, which also summarises the expected behaviour of each signal.

Table 1. Grip and muscle activity experiment: protocol and expected signals.

Time	Grip / Force	Expected EMG
0–20 s	Normal grip (~30–70 N)	Smaller, less dense bursts
25–45 s	50 % grip (~40–100 N)	Denser and taller
50–60 s	Maximum grip (~40–100 N)	Very dense and tall
Rest periods (between grips)	No grip (~0 N)	Almost no signal

Table 2. Maximum, minimum and median force for each student.

Student	Maximum force (N)	Minimum force (N)	Median force (N)
(1) Carolina Martín	174.51	−2.08	37.51
(2) Sandra Rodríguez	171.74	−0.73	12.83

1) Analyse the screenshot of your personal data. Was there any noticeable correlation between muscle activity and grip strength?

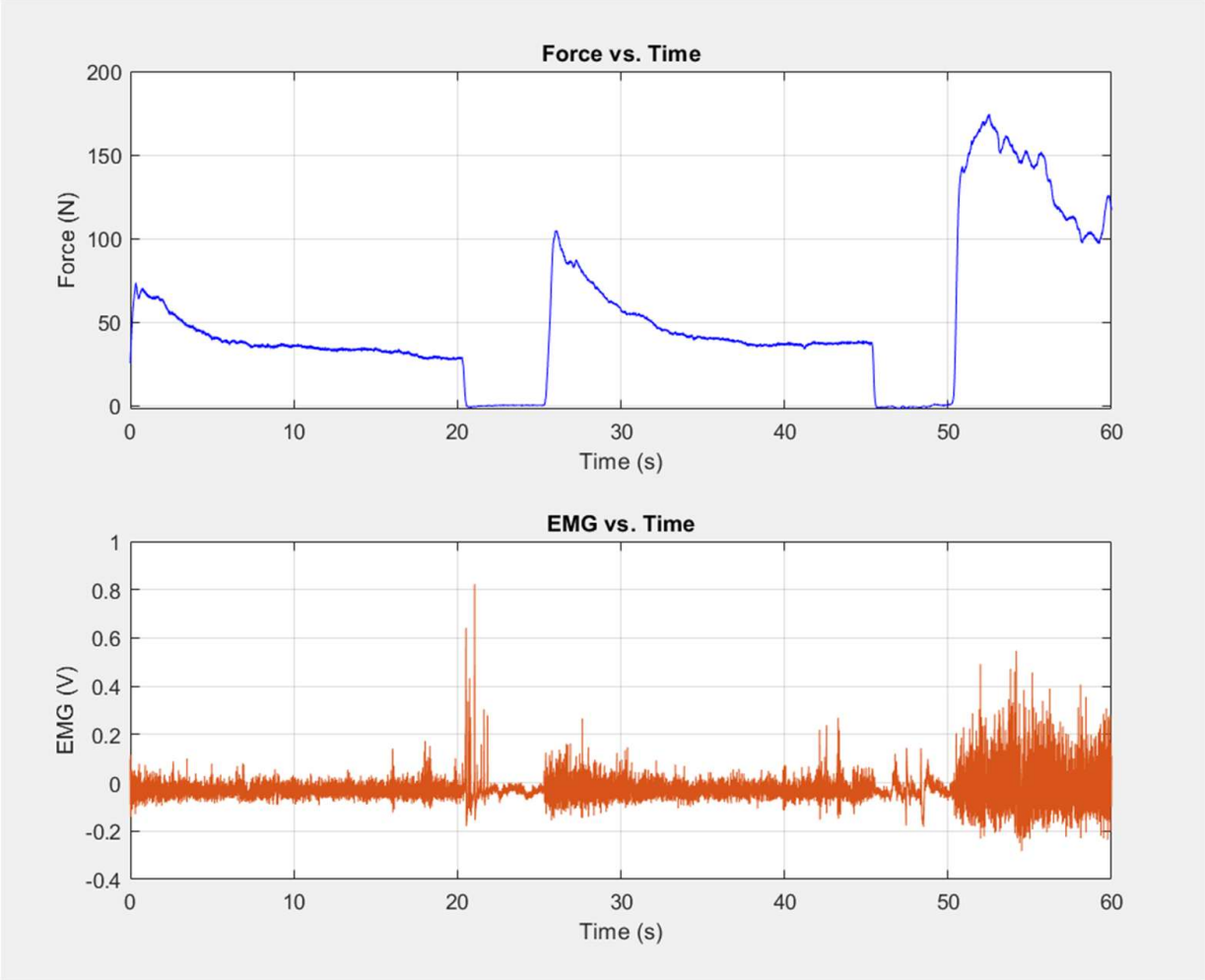


Figure 3. Force and EMG vs Time — Student 1.

The signals from Student 1 show a clear relationship between applied force and EMG amplitude. Comparing the first section (normal grip) with the last section (maximum grip), the EMG is narrower in the first case (approximately between −0.1 V and 0.1 V) and has a much larger amplitude in the last one (approximately between −0.2 V and 0.2–0.4 V). This agrees with the theory: when more force is needed, the nervous system recruits more motor units and increases their firing rate.

However, the EMG signal is often affected by noise and artefacts caused by movement of the electrodes or cables, or by activity of other muscles not directly involved in the grip (e.g. muscles stabilising the arm or maintaining posture). A clear example is the spike at around 21 s in Figure 3: at that moment the force is 0 N, so the EMG should be minimal, but it is not. This spike is therefore most likely a mechanical or cable artefact rather than a muscle event.

2) Compare the screenshot of your personal data with the other members of your group. Does their data support the same trends you recorded in the previous question?

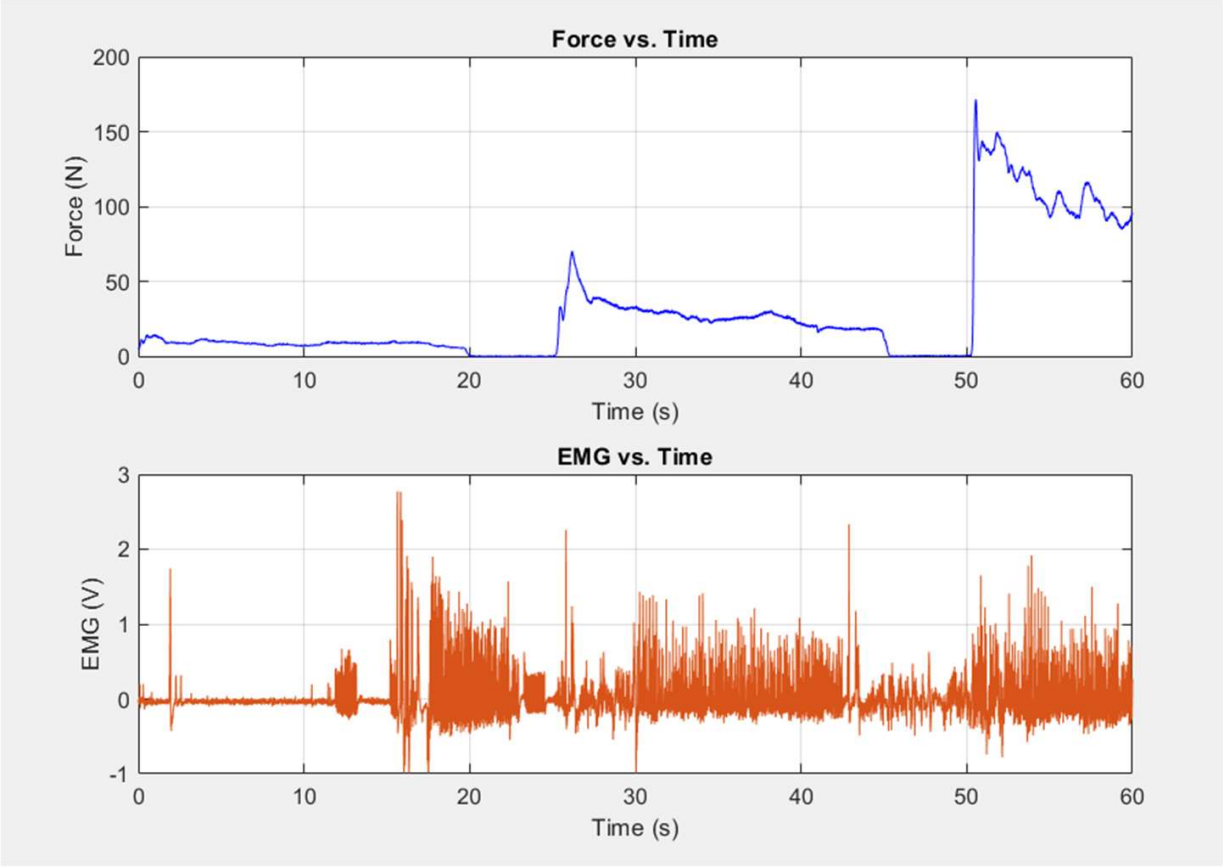


Figure 4. Force and EMG vs Time — Student 2.

For Student 2, during the initial normal-grip period the EMG remains small for most of the first 10 seconds, which again supports a proportional relationship between grip strength and EMG. However, the rest of Student 2's EMG recording is not very reliable: the signal reaches 1–2 V while oscillating, whereas for Student 1 and for the reference recording in the lab script (Figure 2) it stays below 1 V. This suggests cable or electrode artefacts. In addition, the student may not have kept a stable posture or may

have moved her arm even while not squeezing the dynamometer. All of these factors increase the EMG signal even when no grip force is being recorded.

3) Open the saved files with MATLAB and obtain the correlation between EMG amplitude and exerted force. To do so, first rectify the EMG signal and obtain its envelope (a low-pass-filtered version of the signal).

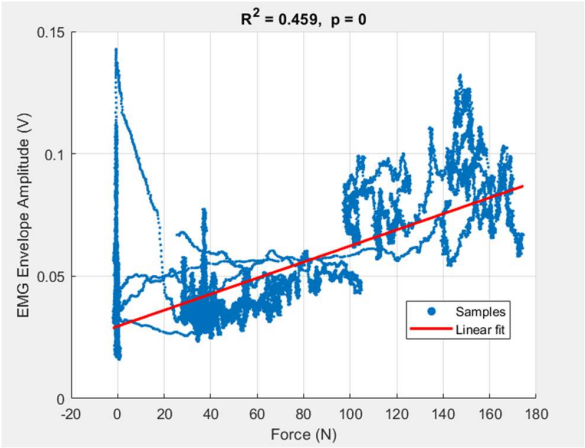


Figure 5. EMG amplitude vs exerted force — Student 1 (R^2 = 0.459).

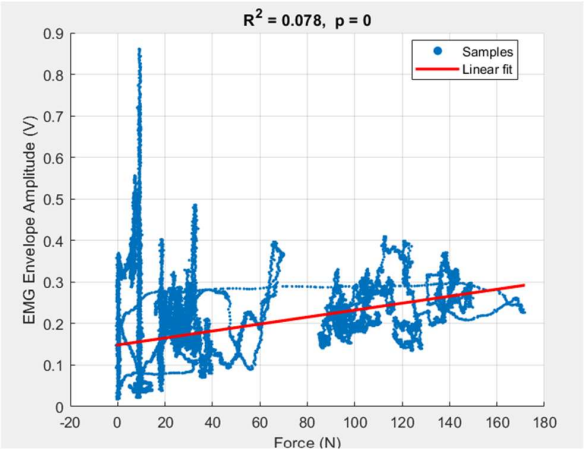


Figure 6. EMG amplitude vs exerted force — Student 2 (R^2 = 0.078).

4) Show the correlation in a graph (ideally with the fitted line). Display the R^2 value and the corresponding p-value. Use adequate axis labels and units.

In both students the red fitted line has a positive slope, showing a positive correlation between EMG envelope amplitude and the force exerted on the dynamometer. The trend is stronger for Student 1, since Student 2's EMG signal was more distorted. Student 1 shows a compact cloud of points between approximately 0.02 and 0.14 V, which rises as more force is applied. Student 2 has a much wider y-axis range (0–0.9 V), with a large concentration of points reaching ~0.85 V when the force is minimal (around 0–20 N). This indicates that noise and artefacts dominate over the muscle signal in Student 2's data.

The near-vertical streak on the left of both plots is probably caused by the transitions into rest: when the force drops to almost 0 N, the EMG envelope takes a moment longer to decay from its previous level (the low-pass filter introduces a lag). This mismatch between the two signals creates the artefact seen in both plots.

The coefficient of determination, R^2, is the fraction of the variance in the EMG amplitude that is explained by the applied force. For noisy biological data such as surface EMG, Student 1's **R^2 = 0.459** indicates a real relationship between both measurements, whereas Student 2's **R^2 = 0.078** indicates almost no relationship and points to a data-acquisition problem. Finally, **p ≈ 0** in both cases because thousands of samples were used; it shows that the correlation is very unlikely to be due to chance, but it says nothing about the *strength* of the relationship, which is better described by R^2 and the slope of the fitted line.

3. Muscle Fatigue

3.1 Data Acquisition

The protocol of each experiment in this section was as follows:

- **Part 2a — Grip intensity (110 s recording):** 20 % strength (20 s) – rest (10 s) – 40 % strength (20 s) – rest (10 s) – 60 % strength (20 s) – rest (10 s) – 80 % strength (20 s).
- **Part 2b — Holding a weight (140 s recording, performed twice: once with a light weight and once with a heavy weight):** hold the weight with the elbow at 90° (40 s) – rest (10 s) – hold (40 s) – rest (10 s) – hold (40 s).
- **Part 2c — Dominant vs non-dominant hand (35 s per hand):** 50 % of maximal grip strength (15 s) – rest (10 s) – maximal grip strength (5 s).

3.1.1 Student 1 — Data

Part 2a. Muscle fatigue at different grip intensities

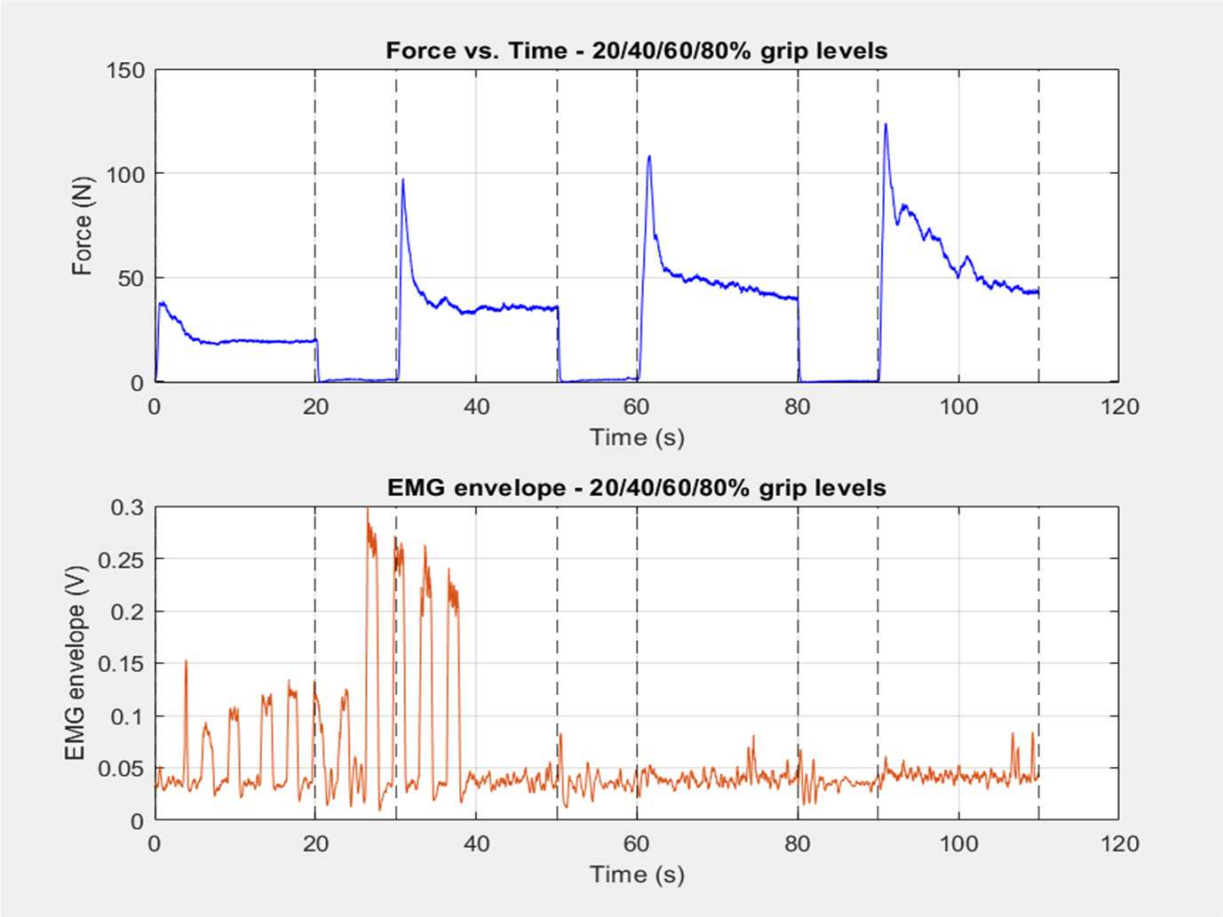


Figure 7. Part 2a, Student 1 — Force and EMG envelope at 20/40/60/80 % grip.

Table 3. Part 2a, Student 1 — Maximum force and mean EMG for each grip level.

Grip level	Max force (N)	Mean EMG (V)
20 %	38.47	0.0556
40 %	97.42	0.0742
60 %	108.60	0.0402
80 %	124.08	0.0433

Part 2b. Muscle fatigue while holding a weight

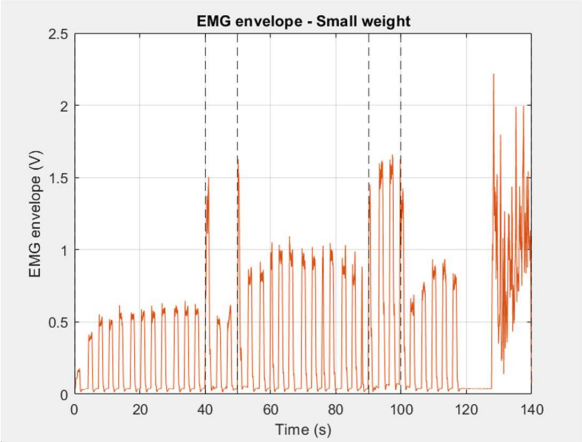


Figure 8. Part 2b, Student 1 — EMG envelope, small weight.

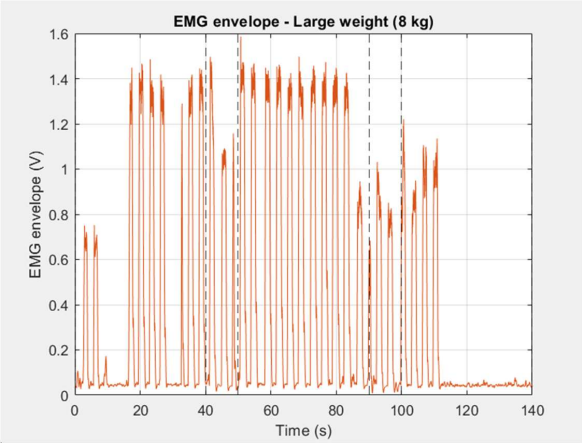


Figure 9. Part 2b, Student 1 — EMG envelope, large weight (8 kg).

Table 4. Part 2b, Student 1 — Mean EMG during each hold.

Hold	Time window	Small weight — mean EMG (V)	Large weight (8 kg) — mean EMG (V)
1	0–40 s	0.2040	0.3476
2	50–90 s	0.3343	0.5791
3	100–140 s	0.2635	0.1647

Part 2c. Dominant vs non-dominant hand

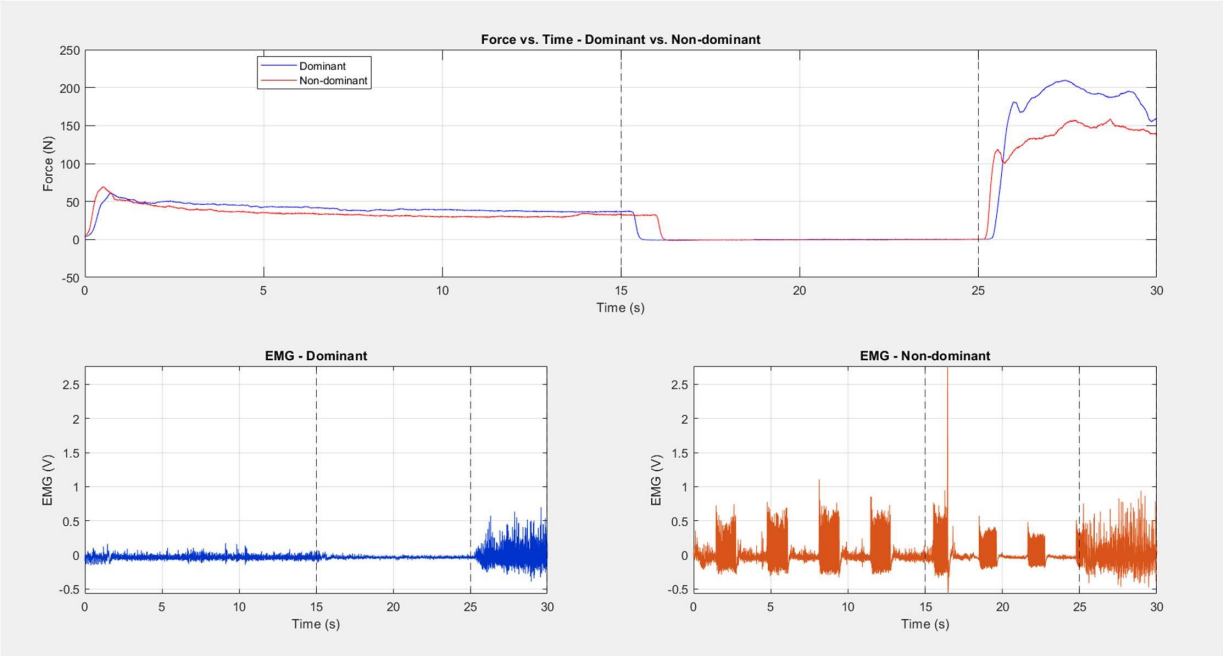


Figure 10. Part 2c, Student 1 — Force and EMG, dominant vs non-dominant hand.

3.1.2 Student 2 — Data

Part 2a. Muscle fatigue at different grip intensities

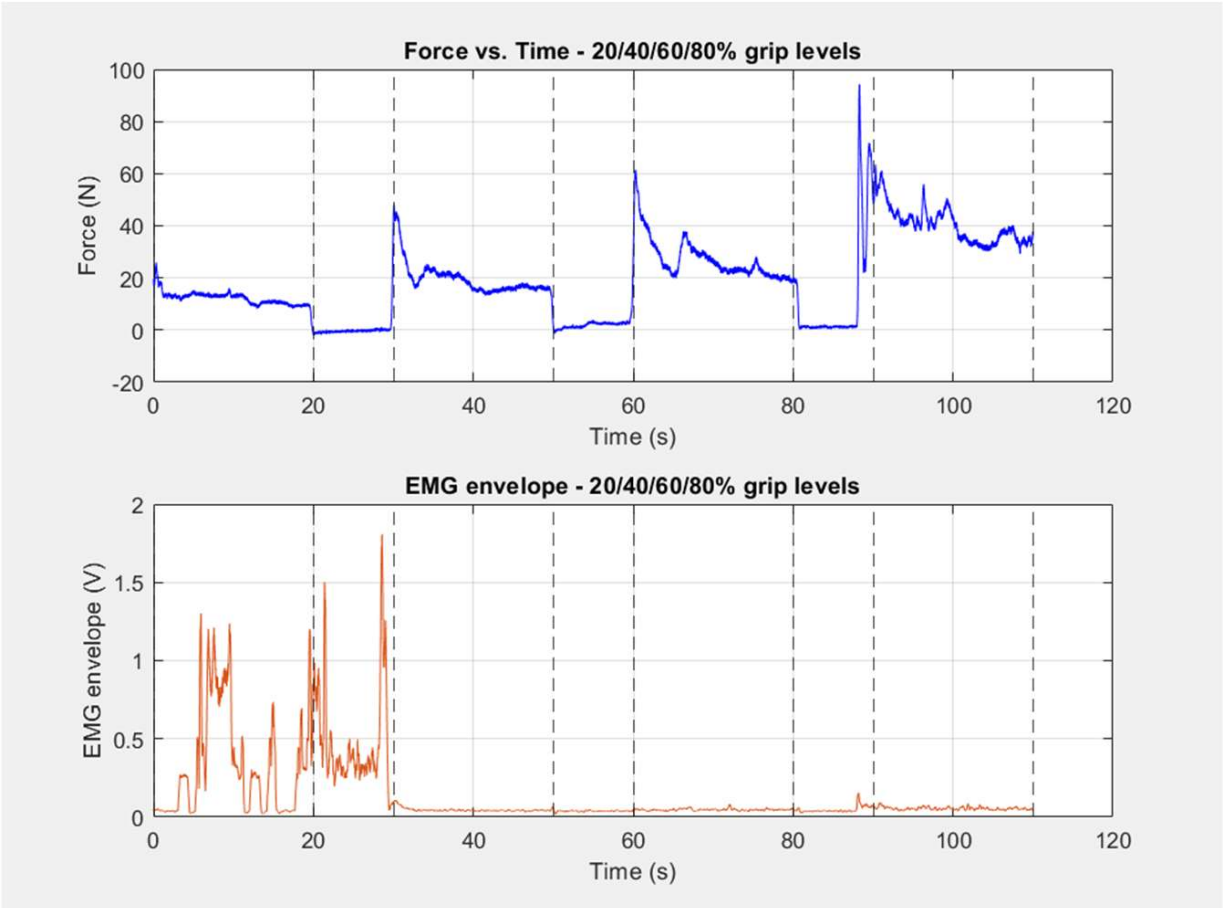


Figure 11. Part 2a, Student 2 — Force and EMG envelope at 20/40/60/80 % grip.

Table 5. Part 2a, Student 2 — Maximum force and mean EMG for each grip level.

Grip level	Max force (N)	Mean EMG (V)
20 %	25.99	0.3317
40 %	48.12	0.0456
60 %	61.45	0.0469
80 %	63.43	0.0546

Part 2b. Muscle fatigue while holding a weight

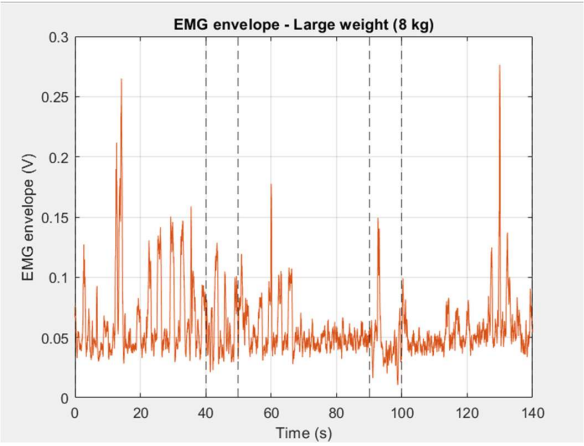


Figure 12. Part 2b, Student 2 — EMG envelope, large weight (8 kg).

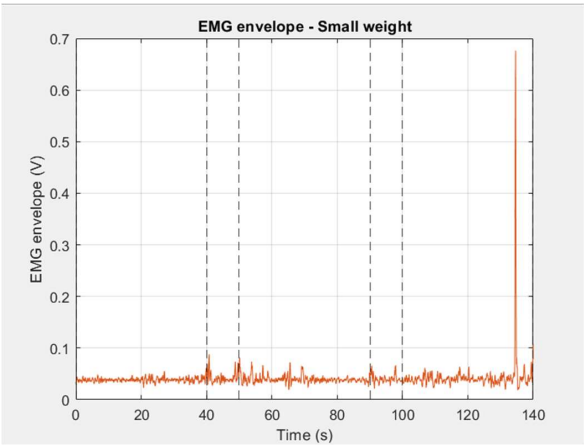


Figure 13. Part 2b, Student 2 — EMG envelope, small weight.

Table 6. Part 2b, Student 2 — Mean EMG during each hold.

Hold	Time window	Small weight — mean EMG (V)	Large weight (8 kg) — mean EMG (V)
1	0–40 s	0.0386	0.0666
2	50–90 s	0.0397	0.0546
3	100–140 s	0.0445	0.0565

3.2 Data Analysis

Table 7. Maximum grip force reached at each target grip level.

Student	20 % (N)	40 % (N)	60 % (N)	80 % (N)
(1) Carolina Martín	38.47	97.42	108.60	124.08
(2) Sandra Rodríguez	25.99	48.12	61.45	63.43

Part 2a — Grip intensity

1) Analyse the screenshot of your personal data.

• Was there a noticeable trend or difference between 20 %, 40 %, 60 % and 80 % of maximal grip strength?

Yes, for force: the maximum force clearly increases with the target grip percentage in both students (Table 7). The mean EMG should also increase with grip intensity; however, the mean EMG values do not show a consistent trend in this part of the experiment. For Student 1 the mean EMG rises from 20 % to 40 % and then decreases at 60 % and 80 % (Table 3). For Student 2 it increases slightly from 40 % to 80 % (0.0456 → 0.0546 V), but the 20 % value (0.3317 V) is clearly dominated by the large artefact bursts visible in the first 30 s of Figure 11 (Table 5).

• Were you able to maintain 20 % for the entire 20 seconds? What about 40 %, 60 % and 80 %?

In both students, the 20 % grip was maintained for almost the entire 20 seconds, as shown in the Force vs Time plots. For the other levels (40 %, 60 % and 80 %), both students show an initial peak, after which the force decreases during the first seconds and then remains fairly stable for the rest of the interval.

• Why do you think you were (or were not) able to maintain the same grip strength at each percentage?

We believe this is due to a poor perception of one's own grip strength combined with muscle-fibre fatigue. Physiologically, this can be explained by the type of motor units recruited (Henneman's size principle). The first motor units to be recruited, used at low grip percentages, are small and composed of slow-twitch (type I) fibres, which are fatigue-resistant. When more force is required, larger motor units with fast-twitch (type II) fibres are recruited: they produce more force but fatigue much faster. This may explain why the students were not able to maintain the higher percentages for long.

Part 2b — Holding a weight

1) and 2) What happened the longer you held the weight? Compare your data with that of the other members of your group. Does their data confirm the trends you found in your own data?

In theory, holding the weight for longer should increase the EMG amplitude, since the nervous system recruits additional motor units (and increases their firing rate) to compensate for fatigue and keep the arm at 90°. In our recordings this trend is difficult to see directly in the EMG plots of Part 2b. However, the mean EMG of each hold (Tables 4 and 6) provides more information:

- **Small weight:** the mean EMG increased from Hold 1 to Hold 2 in both students. For Student 2 it kept increasing in Hold 3 (0.0386 → 0.0397 → 0.0445 V), whereas for Student 1 it decreased slightly in Hold 3 (0.334 → 0.264 V).
- **Large weight:** the trend is less clear. For Student 1 the mean EMG rose sharply from Hold 1 to Hold 2 (0.348 → 0.579 V) and then dropped in Hold 3 (0.165 V); for Student 2 it decreased slightly after Hold 1. The drop in the last hold may be a recording error, or it may reflect that after so much accumulated fatigue the student could no longer sustain the contraction properly.

Moreover, large signals often appear during the rest periods, probably caused by the arm movement when putting the weight down.

Overall analysis

2) (Optional) Did gender appear to make a difference in your data? What about fitness level?

We could not compare male and female data because both participants were female. Regarding fitness level, Student 1 would be expected to have a stronger grip than Student 2, since she regularly practises activities that train the grip, such as strength training and climbing. The force signals do show a tendency for Student 1 to reach higher grip forces. However, it is difficult to compare the EMG signals, since both recordings contain substantial cable artefacts and activity from other muscles.

3) Describe quantitatively and qualitatively how each of the four varying factors affected muscle fatigue.

Factor	Quantitative	Qualitative
1. Grip intensity (20 → 40 → 60 → 80 %)	Student 1 max force: 38 → 97 → 109 → 124 N Student 2 max force: 26 → 48 → 61 → 63 N	Higher intensity → more motor units recruited → more force produced → faster fatigue (the students could not maintain the force), seen as a decay of force at the higher percentages.
2. Weight size	Student 1 mean EMG, Hold 1: 0.2040 V (small) vs 0.3476 V (large) Student 2 mean EMG, Hold 1: 0.0386 V (small) vs 0.0666 V (large)	A heavier load requires more motor units from the start of the contraction → higher EMG.
3. Duration / repetition	Student 1: 0.204 → 0.334 → 0.264 V (small); 0.348 → 0.579 → 0.165 V (large) Student 2: 0.0386 → 0.0397 → 0.0445 V (small); 0.0666 → 0.0546 → 0.0565 V (large)	Repeated effort increases fatigue and therefore the demand for motor units. With enough accumulated demand (large weight), compensation breaks down by Hold 3 (or the drop is an artefact of putting the weight down).
4. Hand dominance	50 % grip: dominant hand ~2–3 N higher than non-dominant Maximal grip: dominant hand ~50 N higher than non-dominant Non-dominant hand shows higher EMG amplitude	The dominant hand produces more force and needs less EMG per newton than the non-dominant hand.

4) Does the data from each member of the group confirm similar trends?

Yes, partially. In Part 2a, both students confirm that a higher grip percentage produces a higher force, as seen in the plots and in the maximum forces (Table 7). The mean EMG, however, did not increase consistently with grip percentage in either student, probably because of the artefacts described above. In

Part 2b, both students showed a higher mean EMG in Hold 2 than in Hold 1 with the small weight (and Student 2 continued increasing in Hold 3), but this trend was not repeated with the large weight.

5) Which factor had the biggest effect on muscle fatigue? Why do you think this is?

The factor with the biggest effect on muscle fatigue was probably the large weight. This can be seen in Student 1's mean EMG: during the first two holds it rose from 0.348 V to 0.579 V, but in Hold 3 it dropped to 0.165 V. This may indicate that, after so much accumulated fatigue, the neuromuscular system could no longer recruit enough motor units to sustain the weight, although a recording artefact cannot be ruled out.

6) Did you notice any trends in the EMG graphs for each of the varying factors? What does this say about the electrical activity in your muscles?

1. **Grip intensity:** the EMG is expected to scale with force, reflecting a greater neural drive to the muscle. This was clearly observed in the first experiment (Section 2; $R^2 = 0.459$ for Student 1), but in Part 2a the mean EMG did not increase consistently with grip percentage.
2. **Weight size:** before fatigue set in, both students showed a higher mean EMG with the heavier weight, i.e. a heavier load requires more motor units to be recruited.
3. **Duration / repetition:** across the three holds, the mean EMG generally increased from Hold 1 to Hold 2, consistent with additional recruitment to compensate for fatigue. With the large weight, the EMG of Student 1 dropped sharply in Hold 3, suggesting that compensation was no longer possible.
4. **Hand dominance:** the non-dominant hand showed a higher overall EMG (larger and denser) despite producing a lower maximal force. This shows that the dominant hand produces force more efficiently, needing to recruit fewer motor units per newton.

7) Do you think electrode placement had any effect on your data? Can you think of a way to increase the accuracy of your data by changing the number and/or placement of the electrodes?

Electrode placement may be one of the reasons why the EMG recordings of Students 1 and 2 had different voltage scales for the same task in the first experiment. Moreover, their correlation between force and EMG was very different ($R^2 = 0.459$ vs 0.078), which could also be explained by different electrode placement.

To increase accuracy, the electrodes should be placed over the belly of the forearm flexor muscles, not over tendons or bony areas, and the inter-electrode distance should be kept constant between subjects. The skin should also be cleaned before placing the electrodes to reduce skin–electrode impedance. One of the most important improvements would be to fix the cables in place so that they do not move and create the sharp artefacts seen in most of the EMG spikes. Finally, a second electrode pair could be added at a different site: comparing both EMG channels would show whether a spike or burst appears in both (muscle activity) or only in one (a local artefact).

8) Are there any sources of error that might have contributed to unexpected trends?

As discussed throughout the report, the main sources of error are:

- **Motion and cable artefacts (mechanical disturbances):** these probably produced the sharp, sudden spikes that appear when they should not, e.g. during rest periods or when putting the weight down.
- **No feedback on the actual force applied:** the students had to estimate their grip percentage and may not have known what percentage of their maximal grip strength they were actually applying.
- **Electrode displacement and degradation:** electrodes may have loosened during the experiment, and the same electrodes were reused for Students 1 and 2, which may have reduced the quality of the EMG signals.
- **Lack of a standardised procedure:** students were sometimes standing and sometimes sitting; the 90° arm position was not carefully measured; and there was no defined way to rest the arm during the rest periods. This may have introduced different artefacts and different levels of fatigue recovery.